

Roll No.

TEC-759

**B. TECH. (ECE-ESR & IOT)
(SEVENTH SEMESTER)**

MID SEMESTER EXAMINATION, Oct., 2022

INTERNET OF THINGS AND ITS APPLICATIONS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Explain the constraints and challenges of wireless sensor network.

10 Marks (CO1)

OR

- (b) What are the characteristics of IoT ? What are functional blocks of IoT ? Explain with the help of a proper diagram.

10 Marks (CO1)

2. (a) What is M2M ? What is M2M value chain ? How can we connect M2M with IoT ?

10 Marks (CO2)

OR

- (b) How can IoT be implemented for Industrial infrastructure ? Explain your example in context of courier service industry.

10 Marks (CO2)

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(2)

3. (a) A sensor is connected with a microcontroller which is having communication protocols to connect with internet. Draw the possible communication through internet of such device. Can we use wireless sensor network topologies with this device ? Explain your answer with the help of proper diagram and logic. 10 Marks (CO1)

OR

- (b) Explain the outline structure of IoT Architecture. Take example of agriculture in support of IoT architecture. 10 Marks (CO2)
4. (a) What is the difference between the IoT and M2M ? Explain with the help of suitable examples. 10 Marks (CO2)

OR

- (b) What is Software Define Network ? Explain with the help of suitable diagram. What are the key elements of SDN ? 10 Marks (CO2)
5. (a) What are the protocols for communication in IoT ? Explain in detail. How does the IoT gateway work ? Explain with the help of a suitable diagram. 10 Marks (CO1)

OR

- (b) Explain with the help of IoT model for 'logistics' by fitting the task of logistics to different layers of IoT. How can the design be commercialized and used for society ? 10 Marks (CO1)